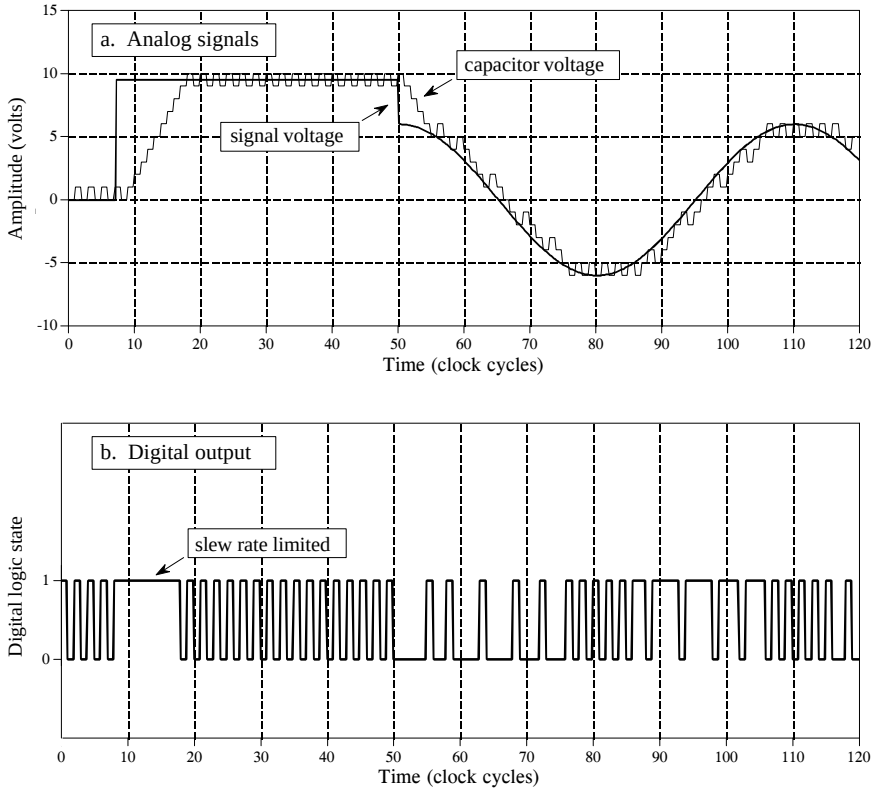

FIGURE 3-17
Example of signals produced by the delta modulator in Fig. 3-16. Figure (a) shows the analog input signal, and the corresponding voltage on the capacitor. Figure (b) shows the delta modulated output, a digital stream of ones and zeros.

spec sheet, not your own work), data rates of 16 to 32 kHz produce acceptable quality speech. The continually changing step size makes the digital data difficult to understand, but fortunately, you don't need to. At the receiver, the analog signal is reconstructed by incorporating a syllabic filter that is identical to the one in the transmission circuit. If the two filters are matched, little distortion results from the CVSD modulation. CVSD is probably the easiest way to digitally transmit a voice signal.

While CVSD modulation is great for encoding voice signals, it cannot be used for general purpose analog-to-digital conversion. Even if you get around the fact that the digital data is related to the *derivative* of the input signal, the *changing step size* will confuse things beyond repair. In addition, the DC level of the analog signal is usually not captured in the digital data.

The **delta-sigma** converter, shown in Fig. 3-19, eliminates these problems by cleverly combining analog electronics with DSP algorithms. Notice that the voltage on the capacitor is now being compared with ground potential. The feedback loop has also been modified so that the voltage on the